Unit 4: Measurement, Drawings, and Manufacturing Documentation

Quiz 4A: Measurement Tools, Precision, Units, and Calipers

Name: ____________________ **Period:** ____________________

Date: ____________________ **Score:** ____________________

Instructions

Answer each question independently. Choose the best answer for multiple choice, complete the matching section, and respond to short-response prompts in complete sentences.

Score:

/ 25

Part A: Multiple Choice

1. Precision refers to...
   A. How exact or repeatable a measurement is
   B. How colorful a drawing is
   C. How many people are on a team
   D. How fast a student works

2. Calipers are useful for measuring...
   A. Small lengths, diameters, depths, and thicknesses
   B. Only time
   C. Only temperature
   D. Only grades

3. Why should units be included with measurements?
   A. Numbers without units can be misunderstood
   B. Units make measurements optional
   C. Units hide errors
   D. Units are only for math class

4. Which is usually more precise?
   A. A ruler with inch marks only
   B. Digital calipers
   C. A guess
   D. A photo

5. What should students do before recording a measurement?
   A. Check the tool and units
   B. Write any number
   C. Copy a partner without checking
   D. Round everything to 10

6. Why measure more than once?
   A. To check consistency and reduce error
   B. To waste time
   C. To avoid tools
   D. To make the packet longer

7. Which measurement is clearest?
   A. 2.5
   B. 2.5 inches
   C. big
   D. about this much

8. Manufacturing depends on measurement because...
   A. Parts must fit and function as intended
   B. Measurements are decorative
   C. Dimensions replace materials
   D. It makes sketches optional

Part B: Vocabulary / Matching

Write the letter of the best matching term in the blank.

_____	**Precision**	A. How exact or repeatable a measurement is.
_____	**Accuracy**	B. How close a measurement is to the true value.
_____	**Caliper**	C. A measuring tool for thickness, diameter, depth, and small lengths.
_____	**Unit**	D. The measurement scale such as inches or millimeters.
_____	**Tolerance**	E. The allowed variation from a target dimension.

Part C: Short Response

1. Explain why a part measurement should include units.

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2. Describe one measurement calipers can make better than a basic ruler.

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Unit 4: Measurement, Drawings, and Manufacturing Documentation

Quiz 4B: Tolerances, Fits, and Manufacturing Notes

Name: ____________ **Period:** ____________

Date: ____________ **Score:** ____________

Instructions

Answer each question independently. Choose the best answer for multiple choice, complete the matching section, and respond to short-response prompts in complete sentences.

Score:

/ 25

Part A: Multiple Choice

1. Tolerance means...
   A. The allowed variation from a target dimension
   B. The final presentation grade
   C. The shape of a logo
   D. The number of sketches

2. Why are tolerances important?
   A. Real manufactured parts are not perfectly exact
   B. They make dimensions unnecessary
   C. They only affect color
   D. They are unrelated to fit

3. A clearance fit means...
   A. Parts have space to move or slide
   B. Parts are glued permanently
   C. Parts never touch
   D. Parts are identical

4. An interference or press fit means...
   A. Parts are intentionally tight
   B. Parts are very loose
   C. Parts are missing
   D. Parts cannot be measured

5. Manufacturing notes may specify...
   A. Material, process, finish, or special instructions
   B. Only jokes
   C. Only class period
   D. Only team name

6. Which is a useful manufacturing note?
   A. Use 3 mm acrylic; laser cut; remove sharp edges
   B. Make it nice
   C. Do something
   D. Looks cool

7. Why should material thickness be measured before laser cutting?
   A. Actual thickness may differ from the listed size
   B. It does not matter
   C. It replaces design work
   D. It only affects color

8. Which factor affects whether two parts fit?

A. Tolerance and manufacturing process
B. Lunch schedule
C. Font choice only
D. The page number

Part B: Vocabulary / Matching

Write the letter of the best matching term in the blank.

____	**Tolerance**	A. The allowed variation from a target measurement.
____	**Clearance fit**	B. A fit with intentional space between parts.
____	**Interference fit**	C. A tight fit where parts press together.
____	**Manufacturing note**	D. Written instruction for making or finishing a part.
____	**Nominal dimension**	E. The target or stated dimension before variation.

Part C: Short Response

1. Explain why a laser-cut slot might need tolerance added to fit a tab.

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2. Write one clear manufacturing note for a 3D printed payload bracket.

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Unit 4: Measurement, Drawings, and Manufacturing Documentation

Quiz 4C: Technical Drawings, BOMs, Revision Control, and Fabrication Planning

Name: ______________________ **Period:** ______________________

Date: ______________________ **Score:** ______________________

Instructions

Answer each question independently. Choose the best answer for multiple choice, complete the matching section, and respond to short-response prompts in complete sentences.

Score:

/ 25

Part A: Multiple Choice

1. A technical drawing communicates...
   A. Shape, size, and manufacturing information
   B. Only opinions
   C. Only group names
   D. Only final grade

2. A BOM is...
   A. Bill of Materials
   B. Best Object Model
   C. Basic Output Method
   D. Bracket Orientation Map

3. A BOM should include...
   A. Part names, quantities, materials, and notes
   B. Only team names
   C. Only sketches
   D. Only colors

4. Revision control tracks...
   A. Changes made to a design or document over time
   B. Who sits where
   C. Only quiz scores
   D. Only material cost

5. Why create a fabrication plan?
   A. To identify steps, tools, materials, and order of operations
   B. To avoid planning
   C. To hide risks
   D. To skip safety

6. A title block often includes...
   A. Part name, scale, author, date, and revision
   B. Only jokes
   C. Only page color
   D. Only final answer

7. Which drawing issue can cause fabrication errors?
   A. Missing or unclear dimensions
   B. Clear notes
   C. A title block
   D. Correct units

8. Why should drawings be checked before making parts?
   A. Mistakes on paper can become wasted material or bad parts
   B. Checking is optional
   C. It slows everyone down only
   D. It replaces cleanup

Part B: Vocabulary / Matching

Write the letter of the best matching term in the blank.

_____	**Technical drawing**	A. A drawing that communicates geometry, dimensions, and manufacturing information.
_____	**BOM**	B. A list of parts, quantities, materials, and notes.
_____	**Revision**	C. A tracked change to a document or design.
_____	**Fabrication plan**	D. A step-by-step plan for making a part or assembly.
_____	**Title block**	E. A drawing area containing identification and documentation information.

Part C: Short Response

1. Explain why a BOM is useful before fabrication starts.

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2. Describe two items that should be included in a fabrication plan.

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